

# FLUXtrapolation Leaderboard

This leaderboard tracks machine-learning model performance on the FLUXtrapolation benchmark across three evaluation settings (temporal split, spatial split, temperature-based split) and three carbon/water flux targets (GPP, ET, NEE). See the [GitHub repository](#) for submission instructions.

ET

GPP

NEE

## Median RMSE

|       |              | Summary                | temporal |        |          |       |        |           | spatial |        |          |       |        |           | temperature |        |          |       |        |           |
|-------|--------------|------------------------|----------|--------|----------|-------|--------|-----------|---------|--------|----------|-------|--------|-----------|-------------|--------|----------|-------|--------|-----------|
|       |              | Skill score $\uparrow$ | hourly   | weekly | seasonal | anom  | iav    | site-mean | hourly  | weekly | seasonal | anom  | iav    | site-mean | hourly      | weekly | seasonal | anom  | iav    | site-mean |
| model | val_strategy |                        |          |        |          |       |        |           |         |        |          |       |        |           |             |        |          |       |        |           |
| coral | mean         | 0.25                   | 0.039    | 0.017  | 0.015    | 0.016 | 0.0025 | 0.0068    | 0.048   | 0.024  | 0.02     | 0.018 | 0.004  | 0.0096    | 0.059       | 0.024  | 0.02     | 0.019 | 0.0037 | 0.0084    |
| mlp   | mean         | 0.24                   | 0.038    | 0.018  | 0.016    | 0.015 | 0.0023 | 0.0069    | 0.048   | 0.024  | 0.018    | 0.018 | 0.0041 | 0.0096    | 0.058       | 0.026  | 0.019    | 0.019 | 0.0043 | 0.0094    |
| xgb   | mean         | 0.23                   | 0.039    | 0.018  | 0.016    | 0.016 | 0.0025 | 0.007     | 0.047   | 0.024  | 0.02     | 0.018 | 0.0042 | 0.011     | 0.055       | 0.024  | 0.019    | 0.019 | 0.0044 | 0.0082    |
| gdro  | mean         | 0.17                   | 0.04     | 0.019  | 0.017    | 0.016 | 0.0025 | 0.0089    | 0.048   | 0.025  | 0.018    | 0.018 | 0.0042 | 0.01      | 0.065       | 0.03   | 0.022    | 0.02  | 0.0039 | 0.017     |
| lr    | mean         | 0.00                   | 0.06     | 0.028  | 0.026    | 0.019 | 0.0025 | 0.012     | 0.061   | 0.029  | 0.026    | 0.021 | 0.0048 | 0.014     | 0.075       | 0.031  | 0.025    | 0.023 | 0.0043 | 0.015     |
| mmd   | mean         | -0.31                  | 0.043    | 0.023  | 0.022    | 0.015 | 0.0026 | 0.014     | 0.094   | 0.055  | 0.053    | 0.024 | 0.0047 | 0.028     | 0.11        | 0.05   | 0.048    | 0.022 | 0.0046 | 0.027     |

## 90th-percentile RMSE

|       |              | Summary                | temporal |        |          |       |        |           | spatial |        |          |       |        |           | temperature |        |          |       |        |           |
|-------|--------------|------------------------|----------|--------|----------|-------|--------|-----------|---------|--------|----------|-------|--------|-----------|-------------|--------|----------|-------|--------|-----------|
|       |              | Skill score $\uparrow$ | hourly   | weekly | seasonal | anom  | iav    | site-mean | hourly  | weekly | seasonal | anom  | iav    | site-mean | hourly      | weekly | seasonal | anom  | iav    | site-mean |
| model | val_strategy |                        |          |        |          |       |        |           |         |        |          |       |        |           |             |        |          |       |        |           |
| coral | mean         | 0.17                   | 0.062    | 0.033  | 0.027    | 0.024 | 0.0077 | 0.021     | 0.074   | 0.039  | 0.035    | 0.026 | 0.0084 | 0.026     | 0.084       | 0.047  | 0.037    | 0.024 | 0.0087 | 0.034     |
| xgb   | mean         | 0.16                   | 0.066    | 0.035  | 0.029    | 0.024 | 0.0084 | 0.022     | 0.075   | 0.037  | 0.033    | 0.026 | 0.0084 | 0.028     | 0.084       | 0.042  | 0.034    | 0.025 | 0.0094 | 0.032     |
| mlp   | mean         | 0.15                   | 0.063    | 0.033  | 0.028    | 0.023 | 0.008  | 0.02      | 0.072   | 0.042  | 0.037    | 0.026 | 0.0084 | 0.026     | 0.084       | 0.05   | 0.043    | 0.024 | 0.0084 | 0.037     |
| gdro  | mean         | 0.04                   | 0.063    | 0.035  | 0.029    | 0.023 | 0.0082 | 0.023     | 0.075   | 0.036  | 0.033    | 0.026 | 0.0083 | 0.022     | 0.1         | 0.073  | 0.07     | 0.026 | 0.0089 | 0.057     |
| lr    | mean         | 0.00                   | 0.081    | 0.043  | 0.042    | 0.026 | 0.0076 | 0.029     | 0.086   | 0.055  | 0.054    | 0.028 | 0.0094 | 0.039     | 0.1         | 0.048  | 0.043    | 0.028 | 0.0088 | 0.038     |
| mmd   | mean         | -0.15                  | 0.066    | 0.036  | 0.031    | 0.024 | 0.0079 | 0.026     | 0.14    | 0.067  | 0.067    | 0.031 | 0.0092 | 0.053     | 0.16        | 0.063  | 0.058    | 0.032 | 0.011  | 0.05      |